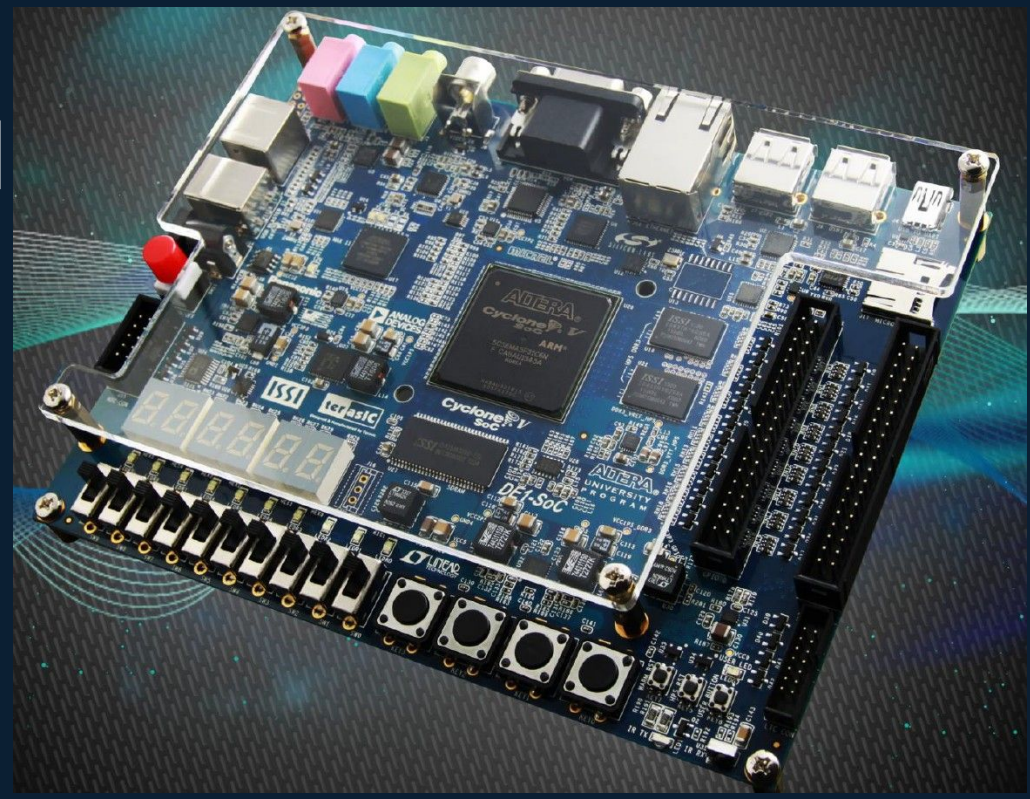
TINYVIT HARDWARE ACCELERATION

TinyViT inference with Hybrid CPU–FPGA on the DE1-SoC



Project 2026272

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02

Motivation and Goal

Motivation

Vision transformers achieve high classification accuracy, but compute-intensive attention and MLP layers pose challenges for resource-constrained edge platforms.

TinyViT Model Selection

21M

parameters

84.8%

ImageNet-1K top-1

Published TinyViT result; board-level performance remains to be measured.

GOAL

Accelerate compute-intensive TinyViT operations on the FPGA while the ARM HPS manages the end-to-end inference pipeline.

Expected Demonstration

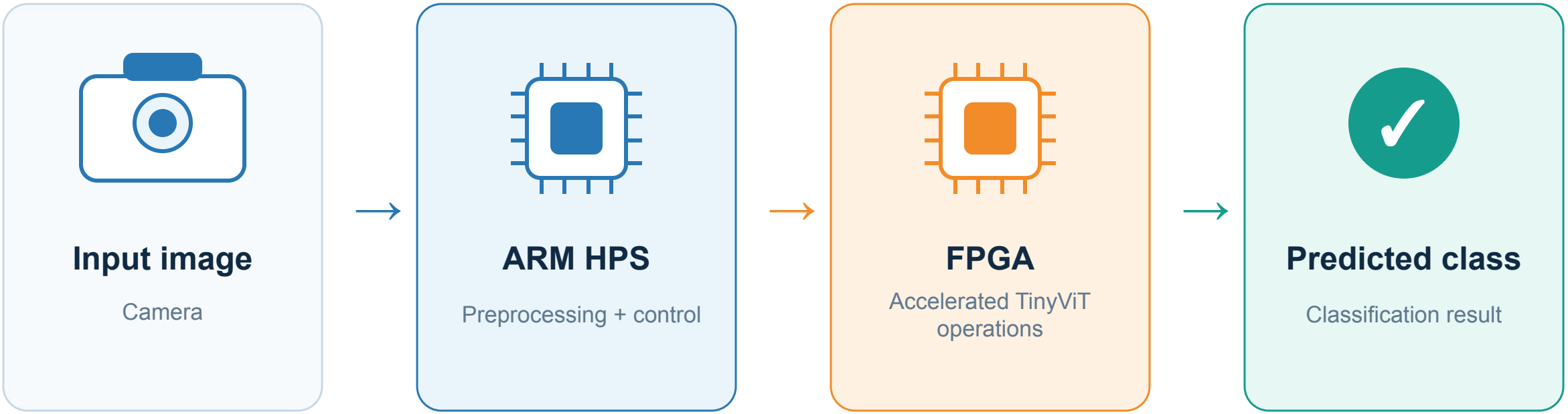
01 **End-to-end image inference**

02 **ARM–FPGA Hardware/Software Integration**

03 **Measured speedup and communication overhead**

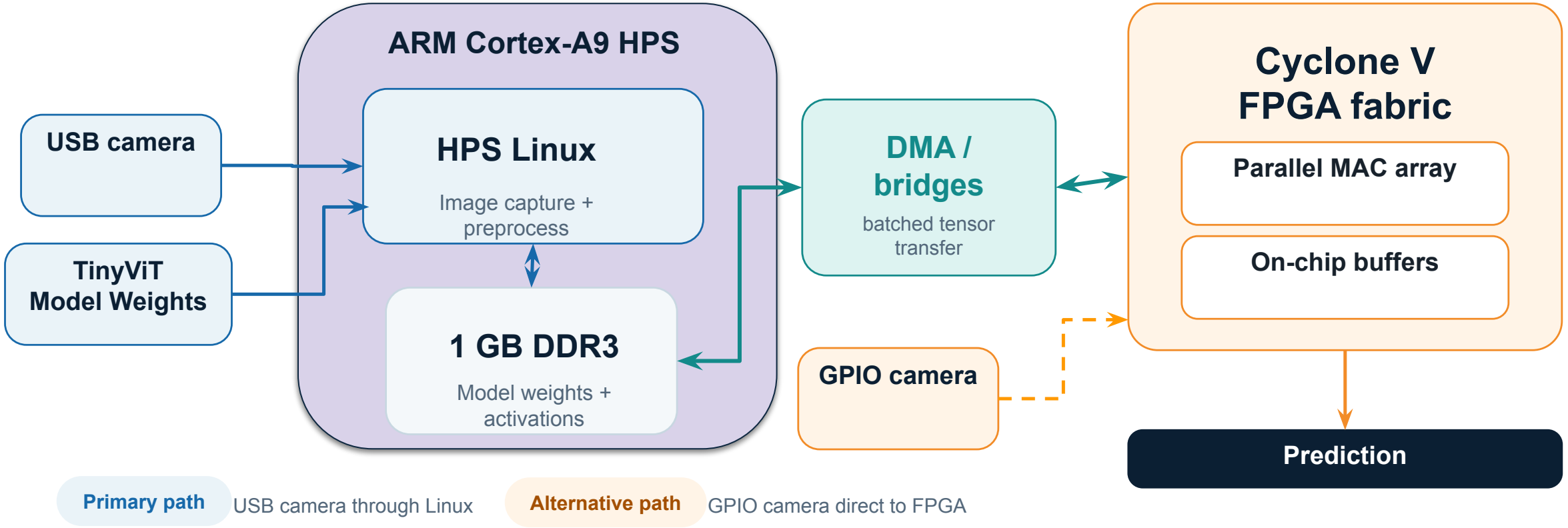
02 Project Description: TinyViT Image Classification on DE1-SoC

Images are processed by the ARM HPS, compute-intensive TinyViT operations are accelerated on the FPGA, and the final classification result is returned to the user.



Hybrid CPU–FPGA inference pipeline

04 System Architecture



HPS–FPGA Workload Partitioning

- **HPS:** Handles control flow and irregular software tasks
- **FPGA:** Accelerates compute-intensive, reusable tensor kernels
- **Data Flow:** HPS orchestrates execution; dense tensor operations are offloaded across the HPS–FPGA bridge

05

FPGA Offload Feasibility

498 ms
CPU GEMM Execution

1.4 ms
DMA Transfer Latency

~54 MACs/B
Arithmetic Intensity

Why Offload Matrix Multiplication?

Compute-Dominated

GEMM execution on the ARM Cortex-A9 takes 498 ms, making computation the primary bottleneck

Low Transfer Overhead

DMA transfer takes only **1.4–6.4** ms, less than **1.3%** of the CPU computation time.

High Arithmetic Intensity

Each byte transferred supports approximately **54 MAC operations**, providing substantial computation per data transfer.

Key Takeaway

Matrix multiplication is a strong FPGA offload candidate: high compute cost and low data-transfer overhead provide significant acceleration potential.

05

Design choices and trade-offs

1

MODEL

TinyViT with INT8 Quantization

TinyViT provides a compact model for edge inference. INT8 quantization reduces the model weights to ~21 MB, while the FPGA provides **low memory budget** of 556 KB embedded on-chip memory.

2

PLATFORM

DE1-SoC

The integrated ARM–FPGA architecture supports hardware/software co-design on a **low-cost**, **familiar** development platform.

3

DEMO TASK

Image Classification

A single predicted class provides a **simple** end-to-end demonstration of the TinyViT inference pipeline.

4

FPGA OFFLOAD

Matrix multiplication

Attention and MLP layers are **dominated** by matrix multiplication, making them well suited for FPGA acceleration. Choosing **systolic array** as acceleration tile for its power and memory efficiency.

07

Implementation and evaluation

Planned Implementation Sequence

1. **CPU baseline — Completed**
   Run TinyViT on the HPS and verify outputs
2. **Profile — Completed**
   Measure layer latency, compute cost, and data-transfer overhead
3. **Accelerate — Sep. 22nd - Dec. 15th**
   Implement and verify the FPGA matrix-multiplication kernel
4. **Integrate — Dec. 15th - Feb. 22nd**
   Integrate DMA, buffering, and HPS–FPGA control
5. **Evaluate — Feb 22nd - End**
   Measure end-to-end performance against the CPU baseline

Success measures

01 End-to-end inference latency
02 FPGA Kernel latency and DMA overhead
03 FPGA logic, DSP, and memory utilization
04 Output consistency with the CPU baseline

Proposed next steps

1. Select numerical precision
2. Define the weight buffering strategy
3. Design systolic array for MatMul acceleration

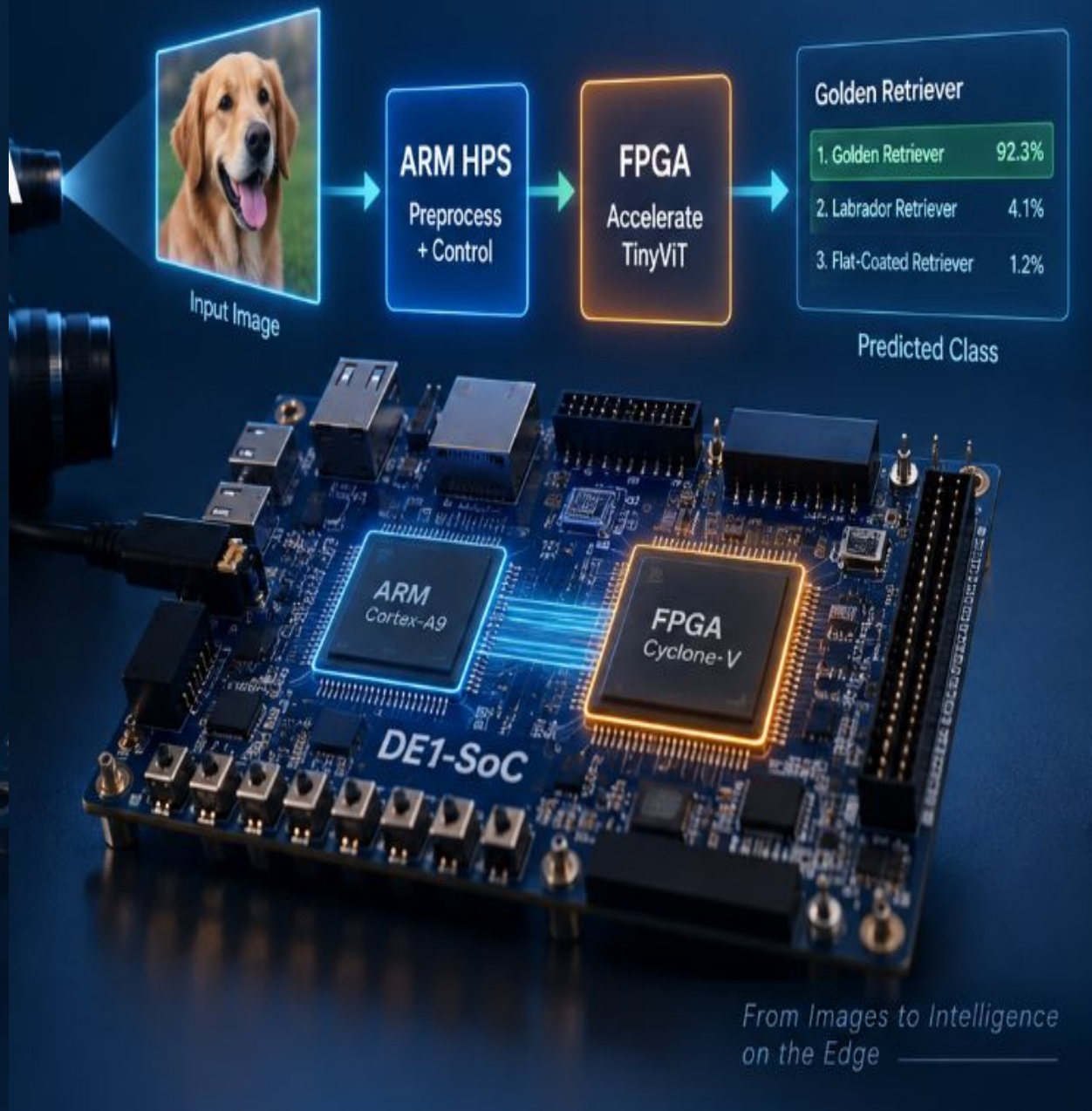
Thank You

Questions & Discussion

TinyViT Hardware Acceleration on the DE1-SoC

Hybrid CPU–FPGA Inference

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From Images to Intelligence on the Edge

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